

CS302-Digital Logic & Design

FINAL TERM Solved MCQS

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Question No: 1

The minimum time for which the input signal has to be maintained at the input of flip-flop is called _____ of the flip-flop.

- ▶ Set-up time
- ▶ **Hold time (Page 242)**
- ▶ Pulse Interval time
- ▶ Pulse Stability time (PST)

Question No: 2

74HC163 has two enable input pins which are _____ and _____

- ▶ **ENP, ENT (Page 285)**
- ▶ ENI, ENC
- ▶ ENP, ENC
- ▶ ENT, ENI

Question No: 3

_____ is said to occur when multiple internal variables change due to change in one input variable

- ▶ Clock Skew
- ▶ **Race condition (Page 267)**
- ▶ Hold delay
- ▶ Hold and Wait

Question No: 4

The _____ input overrides the _____ input

- ▶ **Asynchronous, synchronous (Page 369)**
- ▶ Synchronous, asynchronous
- ▶ Preset input (PRE), Clear input (CLR)
- ▶ Clear input (CLR), Preset input (PRE)

Question No: 5

A decade counter is _____.

- ▶ Mod-3 counter
- ▶ Mod-5 counter
- ▶ Mod-8 counter
- ▶ **Mod-10 counter (Page 274)**

Question No: 6

In asynchronous transmission when the transmission line is idle, _____

- ▶ It is set to logic low
- ▶ **It is set to logic high (Page 356)**
- ▶ Remains in previous state
- ▶ State of transmission line is not used to start transmission

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Question No: 7 (Marks: 1) - Please choose one

A 8-bit serial in / parallel out shift register contains the value “8”, _____ clock signal(s) will be required to shift the value completely out of the register.

- ▶ 1
- ▶ 2
- ▶ 4

▶ 8 (Page 356)

Question No: 8

In a sequential circuit the next state is determined by _____ and _____

- ▶ State variable, current state
- ▶ Current state, flip-flop output

▶ Current state and external input (Page 318)

▶ Input and clock signal applied

Question No: 9

The divide-by-60 counter in digital clock is implemented by using two cascading counters:

▶ Mod-6, Mod-10 (Page 299)

- ▶ Mod-50, Mod-10
- ▶ Mod-10, Mod-50
- ▶ Mod-50, Mod-6

Question No: 10

In NOR gate based S-R latch if both S and R inputs are set to logic 0, the previous output state is maintained.

▶ True (Page 221)

▶ False

Question No: 11

A Nibble consists of _____ bits

▶ 2

▶ 4 (Page 394)

▶ 8

▶ 16

Question No: 12

Excess-8 code assigns _____ to “-8”

▶ 1110

▶ 1100

▶ 1000

▶ 0000 (Page 34)

Question No: 13

The voltage gain of the Inverting Amplifier is given by the relation _____

▶ $V_{out} / V_{in} = - R_f / R_i$ (Page 446)

▶ $V_{out} / R_f = - V_{in} / R_i$

▶ $R_f / V_{in} = - R_i / V_{out}$

▶ $R_f / V_{in} = R_i / V_{out}$

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Question No: 14

LUT is acronym for _____

- ▶ **Look Up Table (Page 439)**
- ▶ Local User Terminal
- ▶ Least Upper Time Period
- ▶ None of given options

Question No: 15

The three fundamental gates are _____

- ▶ AND, NAND, XOR
- ▶ OR, AND, NAND
- ▶ NOT, NOR, XOR
- ▶ **NOT, OR, AND (Page 40)**

Question No: 16

The total amount of memory that is supported by any digital system depends upon _____

- ▶ The organization of memory
- ▶ The structure of memory
- ▶ The size of decoding unit
- ▶ **The size of the address bus of the microprocessor (Page 430)**

Question No: 17

Stack is an acronym for _____

- ▶ FIFO memory
- ▶ **LIFO memory (Page 429)**
- ▶ Flash Memory
- ▶ Bust Flash Memory

Question No: 18

Addition of two octal numbers “36” and “71” results in _____

- ▶ 213
- ▶ 123
- ▶ **127**
- ▶ 345

Question No: 19

_____ is one of the examples of synchronous inputs.

- ▶ **J-K input (Page 235)**
- ▶ EN input
- ▶ Preset input (PRE)
- ▶ Clear Input (CLR)

Question No: 20

_____ occurs when the same clock signal arrives at different times at different clock inputs due to propagation delay.

- ▶ Race condition
- ▶ **Clock Skew (Page 226)**

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- ▶ Ripple Effect
- ▶ None of given options

Question No: 22

In a state diagram, the transition from a current state to the next state is determined by

▶ **Current state and the inputs (Page 332)**

- ▶ Current state and outputs
- ▶ Previous state and inputs
- ▶ Previous state and outputs

Question No: 23

_____ is used to simplify the circuit that determines the next state.

- ▶ State diagram
- ▶ Next state table
- ▶ State reduction
- ▶ **State assignment (Page 335)**

Question No: 24

Assume that a 4-bit serial in/serial out shift register is initially clear. We wish to store the nibble 1100. What will be the 4-bit pattern after the second clock pulse? (Right-most bit first.)

- ▶ 1100
- ▶ 0011
- ▶ 0000
- ▶ 1111

Question No: 25

The operation of J-K flip-flop is similar to that of the SR flip-flop except that the J-K flip-flop _____

▶ **Doesn't have an invalid state (Page 232)**

- ▶ Sets to clear when both $J = 0$ and $K = 0$
- ▶ It does not show transition on change in pulse
- ▶ It does not accept asynchronous inputs

Question No: 26

A multiplexer with a register circuit converts _____

- ▶ Serial data to parallel
- ▶ **Parallel data to serial (Page 356)**
- ▶ Serial data to serial
- ▶ Parallel data to parallel

Question No: 27

GAL is essentially a _____.

- ▶ Non-reprogrammable PAL
- ▶ PAL that is programmed only by the manufacturer
- ▶ Very large PAL
- ▶ **Reprogrammable PAL (Page 183)**

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Question No: 28

in _____, all the columns in the same row are either read or written.

- ▶ Sequential Access
- ▶ MOS Access
- ▶ **FAST Mode Page Access (Page 413)**
- ▶ None of given options

Question No: 29

In order to synchronize two devices that consume and produce data at different rates, we can use _____

- ▶ Read Only Memory
- ▶ **Fist In First Out Memory (Page 425)**
- ▶ Flash Memory
- ▶ Fast Page Access Mode Memory

Question No: 30

A flip-flop changes its state when _____

- ▶ **Low-to-high transition of clock (Page 228)**
- ▶ High-to-low transition of clock
- ▶ Enable input (EN) is set
- ▶ Preset input (PRE) is set

Question No: 31

A frequency counter _____

- ▶ Counts pulse width
- ▶ **Counts no. of clock pulses in 1 second (Page 301)**
- ▶ Counts high and low range of given clock pulse
- ▶ None of given options

Question No: 32

In a sequential circuit the next state is determined by _____ and _____

- ▶ State variable, current state
- ▶ Current state, flip-flop output
- ▶ Current state and external input
- ▶ **Input and clock signal applied (Page 305)**

Question No: 33

Flip flops are also called _____

- ▶ Bi-stable dualvibrators
- ▶ Bi-stable transformer
- ▶ **Bi-stable multivibrators (Page 228)**
- ▶ Bi-stable singlevibrators

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Question No: 36

Given the state diagram of an up/down counter, we can find _____

► **The next state of a given present state (Page 371)**

- The previous state of a given present state
- Both the next and previous states of a given state
- The state diagram shows only the inputs/outputs of a given states

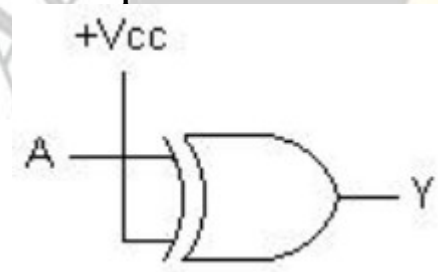
Question No: 38

A Nibble consists of _____ bits

- 2
- **4 (Page 394)**
- 8
- 16

Question No: 39

The output of this circuit is always _____.



- 1
- 0
- **A**
- $\bar{A}$

Question No: 40

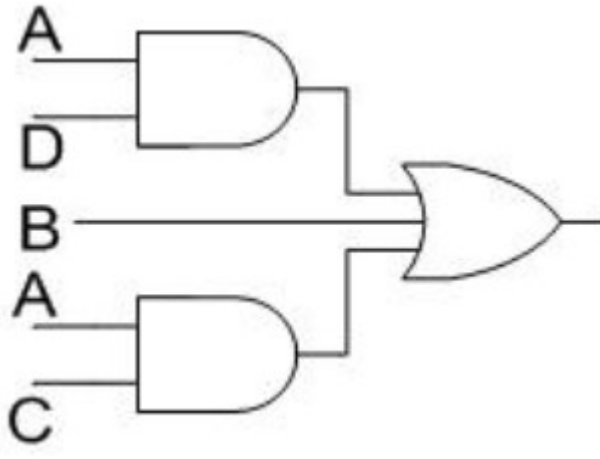
A logic circuit with an output $X = \bar{A}BC + A\bar{B}$ consists of _____.

- two AND gates, two OR gates, two inverters
- three AND gates, two OR gates, one inverter
- **two AND gates, one OR gate, two inverters (Lecture 8)**
- two AND gates, one OR gate

Question No: 41

The diagram given below represents _____

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- ▶ Demorgans law
- ▶ Associative law
- ▶ Product of sum form
- ▶ **Sum of product form (Page 78)**

Question No: 42

The voltage gain of the Inverting Amplifier is given by the relation _____

- ▶ **$V_{out} / V_{in} = - R_f / R_i$ (Page 446)**
- ▶ $V_{out} / R_f = - V_{in} / R_i$
- ▶ $R_f / V_{in} = - R_i / V_{out}$
- ▶ $R_f / V_{in} = R_i / V_{out}$

Question No: 43

DRAM stands for _____

- ▶ **Dynamic RAM (Page 407)**
- ▶ Data RAM
- ▶ Demoduler RAM
- ▶ None of given options

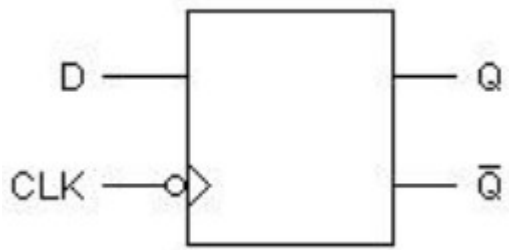
Question No: 44

The three fundamental gates are _____

- ▶ AND, NAND, XOR
- ▶ OR, AND, NAND
- ▶ NOT, NOR, XOR
- ▶ **NOT, OR, AND (Page 40)**

Question No: 45

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Which of the following statement is true regarding above block diagram?

- ▶ Triggering takes place on the negative-going edge of the CLK pulse
- ▶ Triggering takes place on the positive-going edge of the CLK pulse
- ▶ Triggering can take place anytime during the HIGH level of the CLK waveform
- ▶ Triggering can take place anytime during the LOW level of the CLK waveform

Question No: 46

The expression $F=A+B+C$ describes the operation of three bits _____ Gate.

▶ **OR (Page 42)**

- ▶ AND
- ▶ NOT
- ▶ NAND

Question No: 47

Addition of two octal numbers “36” and “71” results in _____

- ▶ 213
- ▶ 123
- ▶ **127**
- ▶ 345

Question No: 48

The ANSI/IEEE Standard 754 defines a _____ Single-Precision Floating Point format for binary numbers.

- ▶ 8-bit
- ▶ 16-bit
- ▶ **32-bit (Page 25)**
- ▶ 64-bit

Question No: 49

The decimal “17” in BCD will be represented as _____

- ▶ 11101
- ▶ 11011
- ▶ **10111 (According to rule)**
- ▶ 11110

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Question No: 50

The basic building block for a logical circuit is _____

- ▶ A Flip-Flop
- ▶ **A Logical Gate (Page 7)**
- ▶ An Adder
- ▶ None of given options

Question No: 51

The output of the expression $F=A.B.C$ will be Logic _____ when $A=1, B=0, C=1$.

- ▶ Undefined
- ▶ One
- ▶ **Zero (According to rule)**
- ▶ No Output as input is invalid.

Question No: 52

_____ is invalid number of cells in a single group formed by the adjacent cells in K-map

- ▶ 2
- ▶ 8
- ▶ **12 (According to rule " 2^n ")**
- ▶ 16

Question No: 53

The PROM consists of a fixed non-programmable _____ Gate array configured as a decoder.

- ▶ **AND (Page 182)**
- ▶ OR
- ▶ NOT
- ▶ XOR

Question No: 54

_____ is one of the examples of asynchronous inputs. ▶ J-K input

- ▶ S-R input
- ▶ D input

▶ Clear Input (CLR) (Page 235)

Question No: 55

Consider an up/down counter that counts between 0 and 15, if external input(X) is "0" the counter counts upward (0000 to 1111) and if external input (X) is "1" the counter counts downward (1111 to 0000), now suppose that the present state is "1100" and $X=1$, the next state of the counter will be

- ▶ 0000
- ▶ **1101 (not sure)** ✗
- ▶ 1011 ✓
- ▶ 1111

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Question No: 56

In a state diagram, the transition from a current state to the next state is determined by

► **Current state and the inputs (Page 232)**

- Current state and outputs
- Previous state and inputs
- Previous state and outputs

Question No: 57

_____ is used to minimize the possible no. of states of a circuit.

► **State assignment (Page 341)**

- State reduction
- Next state table
- State diagram

Question No: 59

The best state assignment tends to _____.

► **Maximizes the number of state variables that don't change in a group of related states (Page 337)**

- Minimizes the number of state variables that don't change in a group of related states
- Minimize the equivalent states
- None of given options

Question No: 60

5-bit Johnson counter sequences through _____ states

- 7
- **10 (Page 354)**
- 32
- 25

Question No: 61

The address from which the data is read, is provided by _____

- Depends on circuitry
- None of given options
- RAM
- **Microprocessor (Page 397)**

Question No: 62

FIFO is an acronym for _____

- **First In, First Out (Page 424)**
- Fly in, Fly Out
- Fast in, Fast Out
- None of given options

Question No: 63

The voltage gain of the Inverting Amplifier is given by the relation _____

- **$V_{out} / V_{in} = - R_f / R_i$ (Page 446)**

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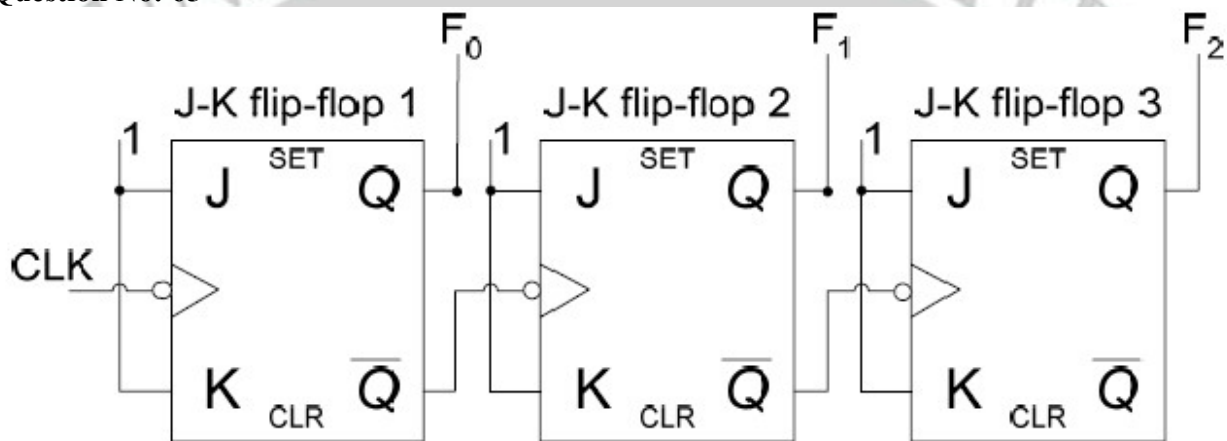
- ▶ $V_{out} / R_f = - V_{in} / R_i$
- ▶ $R_f / V_{in} = - R_i / V_{out}$
- ▶ $R_f / V_{in} = R_i / V_{out}$

Question No: 64

_____ of a D/A converter is determined by comparing the actual output of a D/A converter with the expected output.

- ▶ Resolution
- ▶ **Accuracy (Page 460)**
- ▶ Quantization
- ▶ Missing Code

Question No: 65



Above is the circuit diagram of _____.

- ▶ **Asynchronous up-counter (Page 270)**
- ▶ Asynchronous down-counter
- ▶ Synchronous up-counter
- ▶ Synchronous down-counter

Question No: 66

The sequence of states that are implemented by a n-bit Johnson counter is

- ▶ $n+2$ (n plus 2)
- ▶ **$2n$ (n multiplied by 2) (Page 354)**
- ▶ 2^n (2 raise to power n)
- ▶ n^2 (n raise to power 2)

Question No: 67

" $A + B = B + A$ " is _____

- ▶ Demorgan's Law
- ▶ Distributive Law
- ▶ **Commutative Law (Page 72)**
- ▶ Associative Law

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Question No: 68

Following is standard POS expression

$$(A + \bar{B} + C + \bar{D})(A + \bar{B} + C + D)(A + B + \bar{C} + \bar{D})(A + B + C + \bar{D})(A + \bar{B} + \bar{C} + D)$$

► **True (Lecture 9)**

► False

Question No: 69

An alternate method of implementing Comparators which allows the Comparators to be easily cascaded without the need for extra logic gates is _____

► Using a single comparator

► **Using Iterative Circuit based Comparators (Page 155)**

► Connecting comparators in vertical hierarchy

► Extra logic gates are always required.

Question No: 70

DE multiplexer is also called

► Data selector

► Data router

► **Data distributor (Page 178)**

► Data encoder

Question No: 71

A flip-flop is connected to +5 volts and it draws 5 mA of current during its operation, the power dissipation of the flip-flop is

► 10 mW

► **25 mW (Page 242)**

► 64 mW

► 1024

Question No: 72

Counters as the name indicates are not triggered simultaneously.

► **Asynchronous (Page 269)**

► Synchronous

► Positive-Edge triggered

► Negative-Edge triggered

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Question No: 74

In a state diagram, the transition from a current state to the next state is determined by

► **Current state and the inputs (Page 332)**

- Current state and outputs
- Previous state and inputs
- Previous state and outputs

Question No: 75

A synchronous decade counter will have _____ flip-flops

- 3
- **4 (Page 281)**
- 7
- 10

Question No: 76

The alternate solution for a demultiplexer-register combination circuit is _____

- Parallel in / Serial out shift register
- **Serial in / Parallel out shift register (Page 356)**
- Parallel in / Parallel out shift register
- Serial in / Serial Out shift register

Question No: 77

The 4-bit 2's complement representation of "+5" is _____

- 1010
- 1110
- 1011
- **0101 (Page 22)**

Question No: 78

The storage cell in SRAM is

- a flip-flop
- **a capacitor (Page 407)**
- a fuse
- a magnetic domain

Question No: 79

What is the difference between a D latch and a D flip-flop?

- The D latch has a clock input.
- The D flip-flop has an enable input.
- The D latch is used for faster operation.
- **The D flip-flop has a clock input.**

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Question No: 80

For a positive edge-triggered J-K flip-flop with both J and K HIGH, the outputs Will if the clock goes HIGH.

- ▶ toggle
- ▶ set
- ▶ reset
- ▶ not change

Question No: 81

The OR gate performs Boolean _____.

- ▶ multiplication
- ▶ subtraction
- ▶ division
- ▶ addition (Page 42)

Question No: 82

If an S-R latch has a 1 on the S input and a 0 on the R input and then the S input goes to 0, the latch will be

- ▶ set (Page 219)
- ▶ reset
- ▶ invalid
- ▶ clear

Question No: 83

Determine the values of A, B, C, and D that make the sum term $A(\text{bar}) + B + C(\text{bar}) + D$ equal to zero.

- ▶ A = 1, B = 0, C = 0, D = 0
- ▶ A = 1, B = 0, C = 1, D = 0 (Lecture 8)
- ▶ A = 0, B = 1, C = 0, D = 0
- ▶ A = 1, B = 0, C = 1, D = 1

Question No: 84

The power dissipation, P_D , of a logic gate is the product of the

- ▶ dc supply voltage and the peak current
- ▶ dc supply voltage and the average supply current
- ▶ ac supply voltage and the peak current
- ▶ ac supply voltage and the average supply current

Question No: 85

A Karnaugh map is similar to a truth table because it presents all the possible values of input variables and the resulting output of each value.

- ▶ True
- ▶ False

Question No: 86

NOR Gate can be used to perform the operation of AND, OR and NOT Gate

- ▶ True (Page 50)
- ▶ False

Question No: 87

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Using multiplexer as parallel to serial converter requires _____ connected to the multiplexer

► A parallel to serial converter circuit (Page 244)

- A counter circuit
- A BCD to Decimal decoder
- A 2-to-8 bit decoder

Question No: 88

The 3-variable Karnaugh Map (K-Map) has _____ cells for min or max terms

► 4

► 8 (Page 89)

- 12
- 16

Question No: 89

In designing any counter the transition from a current state to the next state is determined by

► Current state and inputs (Page 332)

- Only inputs
- Only current state
- current state and outputs

Question No: 90

Sum term (Max term) is implemented using _____ gates

► OR (Page 78)

- AND
- NOT
- OR-AND

Question No: 91

AT T_0 THE VALUE STORED IN A 4-BIT LEFT SHIFT WAS "1". WHAT WILL BE THE VALUE OF REGISTER AFTER THREE CLOCK PULSES?

- 2
- 4
- 6

► 8 (not sure)

Question No: 93

If $S=1$ and $R=0$, then $Q(t+1) =$ _____ for positive edge triggered flip-flop

► 0

► 1 (Page 230)

- Invalid
- Input is invalid

Question No: 94

If $S=1$ and $R=1$, then $Q(t+1) =$ _____ for negative edge triggered flip-flop

- 0
- 1

► Invalid (Page 233)

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► Input is invalid

Question No: 95

We have a digital circuit. Different parts of circuit operate at different clock frequencies (4MHZ, 2MHz and 1MHZ), but we have a single clock source having a fix clock frequency (4MHZ), we can get help from

► Using S-R Flop-Flop

► D-flipflop

► **J-K flip-flop (Page 252)**

► T-Flip-Flop

Question No: 96

A counter is implemented using three (3) flip-flops, possibly it will have _____ maximum output status.

► 3

► 7

► **8 (Page 272)**

► 15

Question No: 97

In _____ Q output of the last flip-flop of the shift register is connected to the data input of the first flip-flop of the shift register.

► Moore machine

► Meally machine

► Johnson counter

► **Ring counter (Page 355)**

Question No: 98

The _____ of a ROM is the time it takes for the data to appear at the Data Output of the ROM chip after an address is applied at the address input lines

► Write Time

► Recycle Time

► Refresh Time

► **Access Time (Page 417)**

Question No: 99

Bi-stable devices remain in either of their _____ states unless the inputs force the device to switch its state

► Ten

► Eight

► Three

► **Two (Page 262)**

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Question No: 101

A full-adder has a $C_{in} = 0$. What are the sum (Σ) and the carry (Cout) when $A = 1$ and $B = 1$?

- ▶ $\Sigma = 0$, Cout = 0
- ▶ $\Sigma = 0$, Cout = 1 (Page 135)
- ▶ $\Sigma = 1$, Cout = 0
- ▶ $\Sigma = 1$, Cout = 1

Question No: 102

THE GLITCHES DUE TO RACE CONDITION CAN BE AVOIDED BY USING A _____

- ▶ GATED FLIP-FLOPS
- ▶ PULSE TRIGGERED FLIP-FLOPS
- ▶ POSITIVE-EDGE TRIGGERED FLIP-FLOPS
- ▶ NEGATIVE-EDGE TRIGGERED FLIP-FLOPS (Page 267)

Question No: 103

The design and implementation of synchronous counters start from _____

- ▶ Truth table
- ▶ k-map
- ▶ state table
- ▶ state diagram (Page 319)

Question No: 104

THE HOURS COUNTER IS IMPLEMENTED USING _____

- ▶ ONLY A SINGLE MOD-12 COUNTER IS REQUIRED
- ▶ MOD-10 AND MOD-6 COUNTERS
- ▶ MOD-10 AND MOD-2 COUNTERS
- ▶ A SINGLE DECADE COUNTER AND A FLIP-FLOP (Page 299)

Question No: 105 (Marks: 1) - Please choose one

The high density FLASH memory cell is implemented using _____

- ▶ 1 floating-gate MOS transistor (Page 419)
- ▶ 2 floating-gate MOS transistors
- ▶ 4 floating-gate MOS transistors
- ▶ 6 floating-gate MOS transistors

Question No: 106 (Marks: 1) - Please choose one

$Q2 := Q1 \text{ OR } X \text{ OR } Q3$ The above ABEL expression will be

- ▶ $Q2 := Q1 \$ X \$ Q3$
- ▶ $Q2 := Q1 \# X \# Q3$ (Page 210)
- ▶ $Q2 := Q1 \& X \& Q3$
- ▶ $Q2 := Q1 ! X ! Q3$

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Question No: 107

Generally, the Power dissipation of _____ devices remains constant throughout their operation.

► TTL (Page 65)

- CMOS 3.5 series
- CMOS 5 Series
- Power dissipation of all circuits increases with time.

Question No: 108

When the control line in tri-state buffer is high the buffer operates like a _____ gate

- AND
- OR

► NOT (Page 196)

- XOR

Question No: 109

3.3 v CMOS series is characterized by _____ and _____ as compared to the 5 v CMOS series.

- Low switching speeds, high power dissipation
- Fast switching speeds, high power dissipation
- Fast switching speeds, very low power dissipation (Page 61)
- Low switching speeds, very low power dissipation

Question No: 110

The output of an AND gate is one when _____

► All of the inputs are one (Page 40)

- Any of the input is one
- Any of the input is zero
- All the inputs are zero

Question No: 111

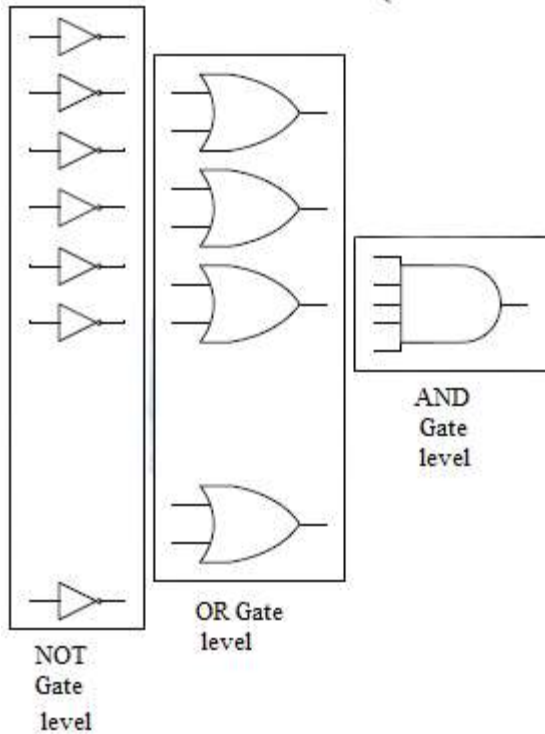
The binary numbers A = 1100 and B = 1001 are applied to the inputs of a comparator. What are the output levels?

- $A > B = 1$, $A < B = 0$, $A = B = 1$
- $A > B = 0$, $A < B = 1$, $A = B = 0$
- $A > B = 1$, $A < B = 0$, $A = B = 0$ (Page 109)
- $A > B = 0$, $A < B = 1$, $A = B = 1$

Question No: 112

The diagram above shows the general implementation of _____ form

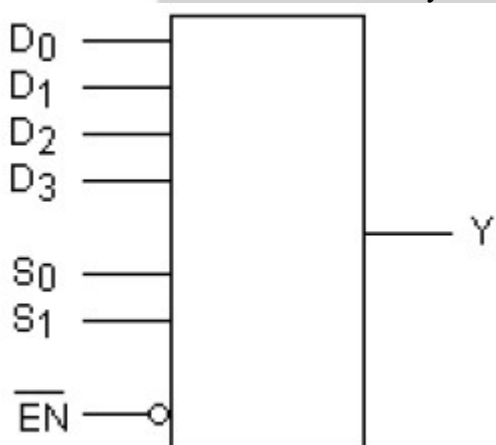
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- ▶ boolean
- ▶ arbitrary
- ▶ **POS (Page 122)**
- ▶ SOP

Question No: 113

The device shown here is most likely a



- ▶ Comparator
- ▶ Multiplexer
- ▶ Demultiplexer
- ▶ Parity generator

Question No: 114

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DE multiplexer converts _____ data to _____ data

- ▶ Parallel data, serial data
- ▶ **Serial data, parallel data (Page 356)**
- ▶ Encoded data, decoded data
- ▶ All of the given options.

Question No:115

If $S=1$ and $R=0$, then $Q(t+1) =$ _____ for positive edge triggered flip-flop

- ▶ 0
- ▶ **1 (Page 230)**
- ▶ Invalid
- ▶ Input is invalid

Question No: 116

If $S=1$ and $R=1$, then $Q(t+1) =$ _____ for negative edge triggered flip-flop

- ▶ 0
- ▶ 1
- ▶ **Invalid (Page 230)**
- ▶ Input is invalid

Question No: 119

In asynchronous digital systems all the circuits change their state with respect to a common clock

- ▶ True
- ▶ **False (Page 245)**

Question No: 201

A flip-flop is connected to +5 volts and it draws 5 mA of current during its operation, the power dissipation of the flip-flop is

- ▶ 10 mW
- ▶ **25 mW (Page 242)**
- ▶ 64 mW
- ▶ 1024

Question No: 202

A divide-by-50 counter divides the input _____ signal to a 1 Hz signal.

- ▶ 10 Hz
- ▶ **50 Hz (Page 298)**
- ▶ 100 Hz
- ▶ 500 Hz

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Question No: 203

The design and implementation of synchronous counters start from _____

- ▶ Truth table
- ▶ k-map
- ▶ state table
- ▶ **state diagram (Page 319)**

Question No: 204

In _____ the output of the last flip-flop of the shift register is connected to the data input of the first flip-flop. ▶ Moore machine

- ▶ Meally machine
- ▶ **Johnson counter (Page 354)**
- ▶ Ring counter Q

Question No: 205

Which is not characteristic of a shift register?

- ▶ **Serial in/parallel in (Page 346)**
- ▶ Serial in/parallel out
- ▶ Parallel in/serial out
- ▶ Parallel in/parallel out

Question No:206

NOR Gate can be used to perform the operation of AND, OR and NOT Gate

- ▶ FALSE
- ▶ **TRUE (Page 250)**

Question No: 207

The output of an XNOR gate is 1 when _____ I) All the inputs are zero II) Any of the inputs is zero III) Any of the inputs is one IV) All the inputs are one

- ▶ I Only
- ▶ IV Only
- ▶ I and IV only
- ▶ **II and III only (Page 53)**

Question No: 208

NAND gate is formed by connecting _____

- ▶ **AND Gate and then NOT Gate (Page 45)**
- ▶ NOT Gate and then AND Gate
- ▶ AND Gate and then OR Gate
- ▶ OR Gate and then AND Gate

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Question No: 209

Consider $A=1, B=0, C=1$. A, B and C represent the input of three bit NAND gate the output of the NAND gate will be _____

► Zero

► **One (Page 46)**

► Undefined

► No output as input is invalid

Question No: 210

The capability that allows the PLDs to be programmed after they have been installed on a circuit board is called _____

► Radiation-Erase programming method (REPM)

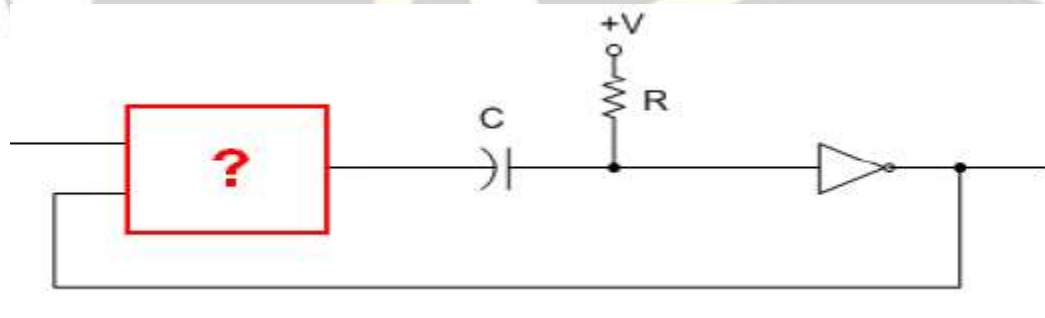
► **In-System Programming (ISP) (Page 194)**

► In-chip Programming (ICP)

► Electronically-Erase programming method (EEPROM)

Question No: 211

Following is the circuit diagram of mono-stable device which gate will be replaced by the red colored rectangle in the circuit.



► AND

► NAND

► NOR

► **XNOR (Page 262)**

Question No: 212

In _____ outputs depend only on the combination of current state and inputs.

► **Mealy machine (Page 332)**

► Moore Machine

► State Reduction table

► State Assignment table

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Question No: 213

In the following statement Z PIN 20 ISTYPE „reg.invert“;

The keyword “reg.invert” indicates _____

- ▶ An inverted register input
- ▶ An inverted register input at pin 20
- ▶ Active-high Registered Mode output
- ▶ **Active-low Registered Mode output (Page 360)**

Question No: 214

A Nibble consists of _____ bits

- ▶ 2
- ▶ **4 (Page 394)**
- ▶ 8
- ▶ 16

Question No: 215

A bidirectional 4-bit shift register is storing the nibble 1110. Its input is LOW. The nibble 0111 is waiting to be entered on the serial data-input line. After two clock pulses, the shift register is storing _____.

- ▶ 1110
- ▶ 0111
- ▶ 1000
- ▶ **1001**

Question No: 216

In order to synchronize two devices that consume and produce data at different rates, we can use _____

- ▶ Read Only Memory
- ▶ **Fist In First Out Memory (Page 425)**
- ▶ Flash Memory
- ▶ Fast Page Access Mode Memory

Question No: 217

If the FIFO Memory output is already filled with data then _____

- ▶ It is locked; no data is allowed to enter
- ▶ It is not locked; the new data overwrites the previous data.
- ▶ Previous data is swapped out of memory and new data enters
- ▶ **None of given options**

Question No: 218

The process of converting the analogue signal into a digital representation (code) is known as _____

- ▶ Strobing
- ▶ Amplification
- ▶ **Quantization (Page 445)**
- ▶ Digitization

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Question No: 219

$$(A + B)(A + \bar{B} + C)(\bar{A} + C)$$

is an example of

► **Product of sum form (Page 77)**

- Sum of product form
- Demorgans law
- Associative law

Question No: 220

Q2 := Q1 OR X OR Q3 The above ABEL expression will be

► Q2 := Q1 \$ X \$ Q3

► **Q2 := Q1 # X # Q3 (Page 210)**

► Q2 := Q1 & X & Q3

► Q2 := Q1 ! X ! Q3

Question No: 221

Caveman number system is Base _____ number system ► 2

► **5 (Page 11)**

- 10
- 16

Question No: 222

The output of an XOR gate is zero (0) when _____

- I) All the inputs are zero
- II) Any of the inputs is zero
- III) Any of the inputs is one
- IV) All the inputs are one

- I Only
- IV Only

► **I and IV only (Page 53)**

- II and III only

Question No: 223

The simplest and most commonly used Decoders are the _____ Decoders

► **n to 2n (Page 158)**

- (n-1) to 2n
- (n-1) to (2n-1)
- n to 2n-1

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Question No: 224

The _____ Encoder is used as a keypad encoder.

- ▶ 2-to-8 encoder
- ▶ 4-to-16 encoder
- ▶ BCD-to-Decimal

▶ **Decimal-to-BCD Priority (Page 166)**

Question No: 225

3-to-8 decoder can be used to implement Standard SOP and POS Boolean expressions

▶ **True (Page 161)**

▶ False

Question No: 226

If $S=1$ and $R=0$, then $Q(t+1) = \underline{\hspace{2cm}}$ for positive edge triggered flip-flop

▶ 0

▶ **1 (Page 230)**

▶ Invalid

▶ Input is invalid

Question No: 227

If the S and R inputs of the gated S-R latch are connected together using a _____ gate then there is only a single input to the latch. The input is represented by D instead of S or R (A gated D-Latch)

▶ AND

▶ OR

▶ **NOT (Page 226)**

▶ XOR

Question No: 228

The low to high or high to low transition of the clock is considered to be a(n) _____

▶ State

▶ **Edge (Page 228)**

▶ Trigger

▶ One-shot

Question No: 230

RCO Stands for _____

▶ Reconfiguration Counter Output

▶ Reconfiguration Clock Output

▶ Ripple Counter Output

▶ **Ripple Clock Output (Page 285)**

Question No: 231

A transparent mode means _____

▶ **The changes in the data at the inputs of the latch are seen at the output (Page 245)**

▶ The changes in the data at the inputs of the latch are not seen at the output

▶ Propagation Delay is zero (Output is immediately changed when clock signal is applied)

▶ Input Hold time is zero (no need to maintain input after clock transition)

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Question No: 232

In _____ outputs depend only on the current state.

- ▶ Mealy machine
- ▶ **Moore Machine (Page 332)**
- ▶ State Reduction table
- ▶ State Assignment table

Question No: 233

Smallest unit of binary data is a _____

- ▶ **Bit (Page 394)**
- ▶ Nibble
- ▶ Byte
- ▶ Word

Question No: 234

NOR gate is formed by connecting _____

- ▶ **OR Gate and then NOT Gate (Page 47)**
- ▶ NOT Gate and then OR Gate
- ▶ AND Gate and then OR Gate
- ▶ OR Gate and then AND Gate

Question No: 235

A particular half adder has

- ▶ 2 INPUTS AND 1 OUTPUT
- ▶ **2 INPUTS AND 2 OUTPUT (Page 134)**
- ▶ 3 INPUTS AND 1 OUTPUT
- ▶ 3 INPUTS AND 2 OUTPUT

Question No: 236

THE FOUR OUTPUTS OF TWO 4-INPUT MULTIPLEXERS, CONNECTED TO FORM A 16-INPUT MULTIPLEXER, ARE CONNECTED TOGETHER THROUGH A 4-INPUT _____ GATE

- ▶ AND
- ▶ **OR (Page 171)**
- ▶ NAND
- ▶ XOR

Question No: 237

A FIELD-PROGRAMMABLE LOGIC ARRAY CAN BE PROGRAMMED BY THE USER AND NOT BY THE MANUFACTURER.

- ▶ **TRUE (Page 182)**
- ▶ FALSE

Question No: 241

A synchronous decade counter will have _____ flip-flops

- ▶ 3
- ▶ **4 (Page 281)**
- ▶ 7

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► 10

Question No: 242

The alternate solution for a multiplexer and a register circuit is _____

► **Parallel in / Serial out shift register (Page 356)**

- Serial in / Parallel out shift register
- Parallel in / Parallel out shift register
- Serial in / Serial Out shift register

Question No: 243

A 8-bit serial in / parallel out shift register contains the value “8”, _____ clock signal(s) will be required to shift the value completely out of the register.

- 1
- 2
- 4

► **8 (Page 356)**

Question No: 245

If the FIFO Memory output is already filled with data then _____

- It is locked; no data is allowed to enter
- It is not locked; the new data overwrites the previous data.
- Previous data is swapped out of memory and new data enters

► **None of given options**

Question No: 246

WHEN BOTH THE INPUTS OF EDGE-TRIGGERED J-K FLOP-FLOP ARE SET TO LOGIC ZERO -----

- THE FLOP-FLOP IS TRIGGERED
- $Q=0$ AND $Q'=1$
- $Q=1$ AND $Q'=0$

► **THE OUTPUT OF FLIP-FLOP REMAINS UNCHANGED (page 223)**

Question No: 247

The terminal count of a 4-bit binary counter in the UP mode is _____.

- 1100
- 0011
- **1111**
- 0000

Question No: 248

For a down counter that counts from (111 to 000). If current state is “101” the next state will be _____.

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- 111
- 110
- 010
- None of given options

Question No: 249

The n flip-flops store _____ states.

- a. 1
- b. 2^n
- c. 0
- d. $2^{(n+1)}$

Question No: 250

An Asynchronous Down-counter is implemented (using J-K flip-flop) by connecting _____.

- Q output of all flip-flops to clock input of next flip-flops
- Q' output of all flip-flops to clock input of next flip-flops
- Q output of all flip-flops to J input of next flip-flops
- Q' output of all flip-flops to K input of next flip-flops

Question No: 251

In case of cascading Integrated Circuit counters, the enable inputs and RCO of the Integrated Circuit counters allow cascading of multiple counters together.

- True
- False

Question No: 251

A decade counter can be implemented by truncating the counting sequence of a MOD-20 counter.

- a. True

- b. False

Question No: 252

The 74HC163 is a 4-bit Synchronous Counter, it has _____ data output pins.

- b. 2
- b. 4
- c. 6
- d. 8

Question No: 253

Divide-by-32 counter can be achieved by using _____.

- c. Flip-Flop and DIV10
- b. Flip-Flop and DIV 16
- c. Flip-Flop and DIV 32
- d. DIV 16 and DIV 32

Question No: 254

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The synchronous counters are also known as Ripple Counters:

a. True

b. False

Question No: 255

Each stage of Master-slave flip-flop works at _____ of the clock signal

- Each stage works on complete clock signal
- One fourth
- One third
- One half

Question No: 256

With a 100 KHz clock frequency, eight bits can be serially entered into a shift register in

- a. 80 micro seconds
- b. 8 micro seconds
- c. 80 mili seconds
- d. 10 micro seconds

Question No: 257

Number of states in an 8-bit Johnson counter sequence are:

- d. 8
- e. 12
- f. 14
- g. 16

Question No: 258

In moore machine the output depends on

- The current state and the output of previous flip flop
- Only inputs
- The current state
- The current state and inputs

Question No: 259

Asynchronous mean that _____

- Each flip-flop after the first one is enabled by the output of the preceding flip-flop
- Each flip-flop is enabled by the output of the preceding flip-flop
- Each flip-flop except the last one is enabled by the output of the preceding flip-flop
- Each alternative flip-flop after the first one is enabled by the output of the preceding flip-flop

Question No: 260

According to moore circuit, the output of synchronous sequential circuit depend/s on _____ of flip flop.

- h. Previous state
- i. Present state
- j. Next state
- k. External state

Question No: 261

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In gated SR latch, what is the value of the output if EN=1, S=0 and R=1?

- ☐ Q
- ☐ 0
- 1
- Invalid

Question No: 262

A Divide-by-20 counter can be achieved by using

- a. Flip-Flop and DIV 10
- b. Flip-Flop and DIV 16
- c. Flip-Flop and DIV 32
- d. DIV 10 and DIV 16

Question No: 263

A one-shot mono-stable device contains_____

- ☐ AND gate, Resistor, Capacitor and NOT Gate
- ☐ NAND gate, Resistor, Capacitor and NOT Gate
- ☐ NOR gate, Resistor, Capacitor and NOT Gate
- XNOR gate, Resistor, Capacitor and NOT Gate

Question No: 264

The inputs can be directly mapped to karnaugh maps.

- S-R
- J-K
- Flip-Flop
- External

Question No: 265

A mono-stable device only has a single stable state

- a. True
- b. False

Question No: 266

When the_____Hz sampling interval is selected, the signal at the output of the J-K flip-flop has a time period of_____seconds.

- a. 1, 2
- b. 0, 2
- c. 2, 5
- d. 1, 1

Question No: 267

Assume a J-K flip-flop has 1s on the J and K inputs. The next clock pulse will cause the output to_____.

- Set
- Toggle

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- Latch
- Reset

Question No: 268

A stage in the shift register consists of

- A latch
- A flip flop
- A byte of storage
- Four bits of storage

Question No: 269

If a circuit suffers “Clock Skew” problem, the output of circuit can’t be guaranteed.

- a. True
- b. False

Question No: 270

A modulus-14 counter has fourteen states requiring _____

- l. 14 flip flops
- m. 14 registers
- c. 4 flip flops
- d. 4 registers

Question No: 271

In Master-Slave flip-flop the clock signal is connected to slave flip-flop using _____ gate.

- AND
- OR
- NOT
- NAND

Question No: 272

_____ flip-flops are obsolete now.

- Edge-triggered
- Master-Slave
- T-flipflop
- D-flipflop

Question No: 273

The glitches due to “Race Condition” can be avoided by using a _____.

- n. Gated flip-flops
- o. Pulse triggered flip-flops
- p. Positive-Edge triggered flip-flops
- d. Negative-Edge triggered flip-flops

Question No: 274

For a gated D-Latch if $EN=1$ and $D=1$ then $Q(t+1)=$ _____

- 0
- $Q(t)$

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- 1

- Invalid

Question No: 275

_____ occurs when the same clock signal arrives at different times at different clock inputs due to propagation delay.

- q. Race condition
- b. **Clock skew**
- c. Ripple effect
- d. None of given options

Question No: 276

An Astable multivibrator is known as a (n)_____.

- a. **Oscillator**
- b. Booster
- c. One-shot
- d. Dual-shot

Question No: 277

In Master-Slave flip-flop setup, the master flip-flop operators at _____

- Positive half cycle of pulse
- Negative half cycle of pulse
- **Both Master-Slave operator simultaneously**
- Master-Slave flip-flop does not operate on pulses rather it is edge triggered

Question No: 278

The power consumed by a flip-flop is defined by _____

- $P = I_{cc} \times R_{cc}$
- $P = V_{cc} \times R_{cc}$
- **c. $P = V_{cc} \times I_{cc}$**
- $P = M_{cc} \times V_{cc}$

Question No: 279

The 3-bit up counter can be implemented using _____ flip-flop(s).

- S-R flip-flops only
- **S-R flip-flops and D-flip-flops**
- S-R flip-flops or D-flip-flops
- D-flip-flop only

Question No: 280

The terminal count of a 4-bit binary counter in the DOWN mode is _____

- a. **0000**
- b. 0011
- c. 1100
- d. 1111

Question No: 281

Which mechanisms allocate the binary values to the states in order to reduce the cost of the combinational circuits?

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- State reduction
- State minimization

State assignment

- State evaluation

Question No: 282

State of flip-flop can be switched by changing its _____

- a. Input signal
- b. Output signal
- c. Momentary signal
- d. Contemporary signal

Question No: 283

Once the state diagram is drawn for any sequential circuit the next step is to draw

- Transition table
- Karnaugh map
- Next-state table
- Logic expression

Question No: 284

Design of state diagram is one of many steps used to design

- A clock
- A truncated counter
- An UP/DOWN counter
- Any counter

Question No: 285

Flip flops are also called _____

- Bi-stable multivibrators
- Bi-stable single vibrators
- Bi-stable dual vibrators
- Bi-stable transformer

Question No: 286

Three cascaded modulus-10 counters have an overall modulus of

- 30
- b. 100
- c. 1000
- 10000

Question No: 287

The term hold always means _____.

- a. $Q=0, Q'=1$
- b. $Q=1, Q'=0$
- c. $Q=0, Q'=0$
- d. No change

Question No: 288

A flip-flop is presently in SET state and must remain SET on the next clock pulse. What must j and k be?

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- a. $J=1, K=0$
- b. $J=1, K=X$ (Don't care)
- c. $J=X$ (Don't care), $K=0$
- d. $J=0, K=X$ (Don't care)

Question No: 289

To parallel load a byte of data into a shift register, there must be

- a. One clock pulse
- b. One clock pulse for each 1 in the data
- c. Eight clock pulse
- d. One clock pulse for each 0 in the data

Question No: 290

Invalid state of NOR based SR latch occurs when _____.

- r. $S=0, R=0$
- s. $S=0, R=1$
- t. $S=1, R=0$
- u. $S=1, R=1$

